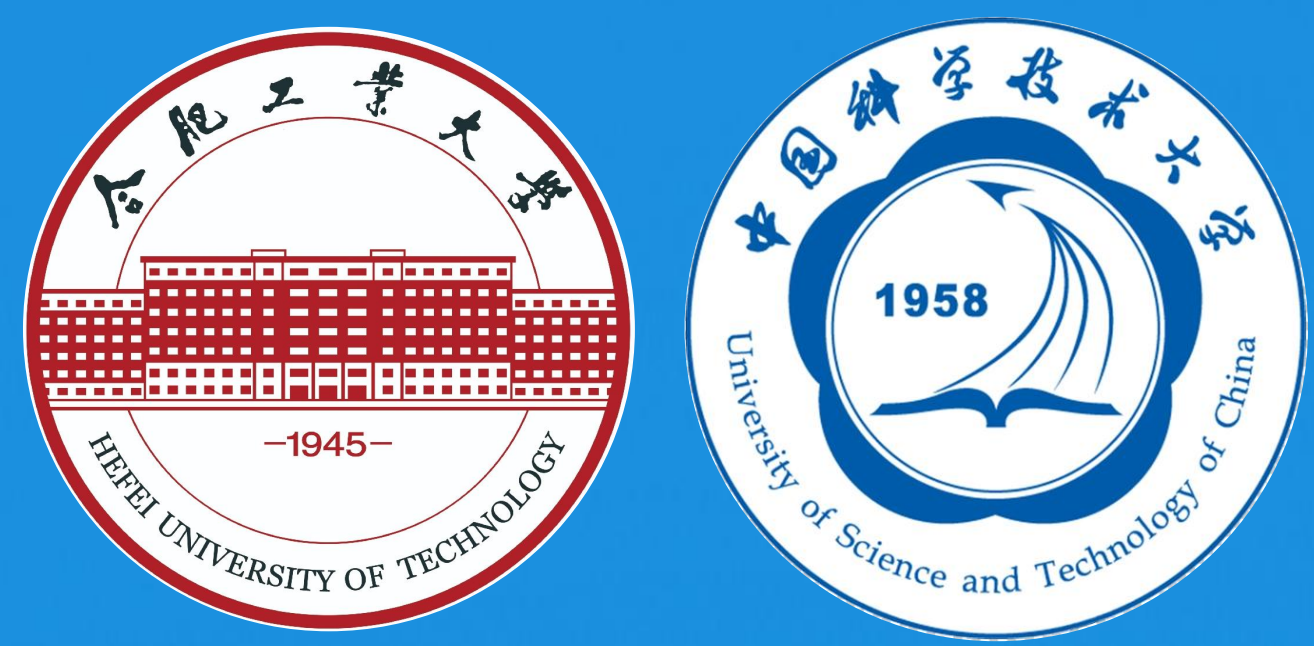




Unified Static and Dynamic Network: Efficient Temporal Filtering for Video Grounding

Jingjing Hu¹, Dan Guo¹, Kun Li¹, Zhan Si², Xun Yang²,
Xiaojun Chang², and Meng Wang (*IEEE Fellow*)¹
¹HFUT, ²USTC

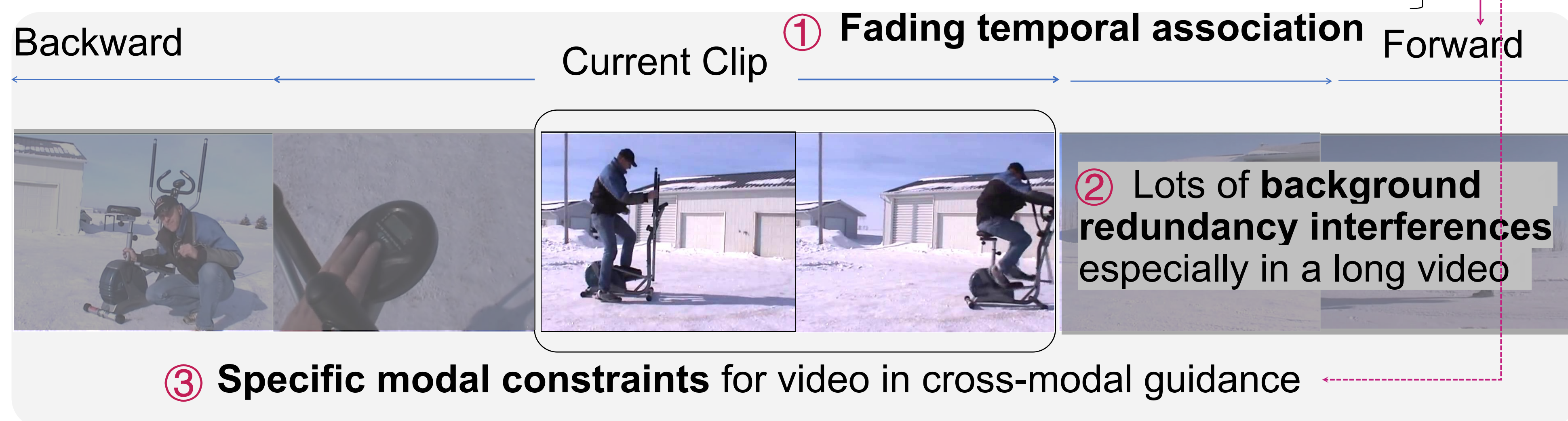
<https://github.com/xian-sh/UniSDNet>

BackGround

Task: Given an untrimmed video and a query (text or audio), the task is to identify the temporal moment(s) whose semantics match the query.

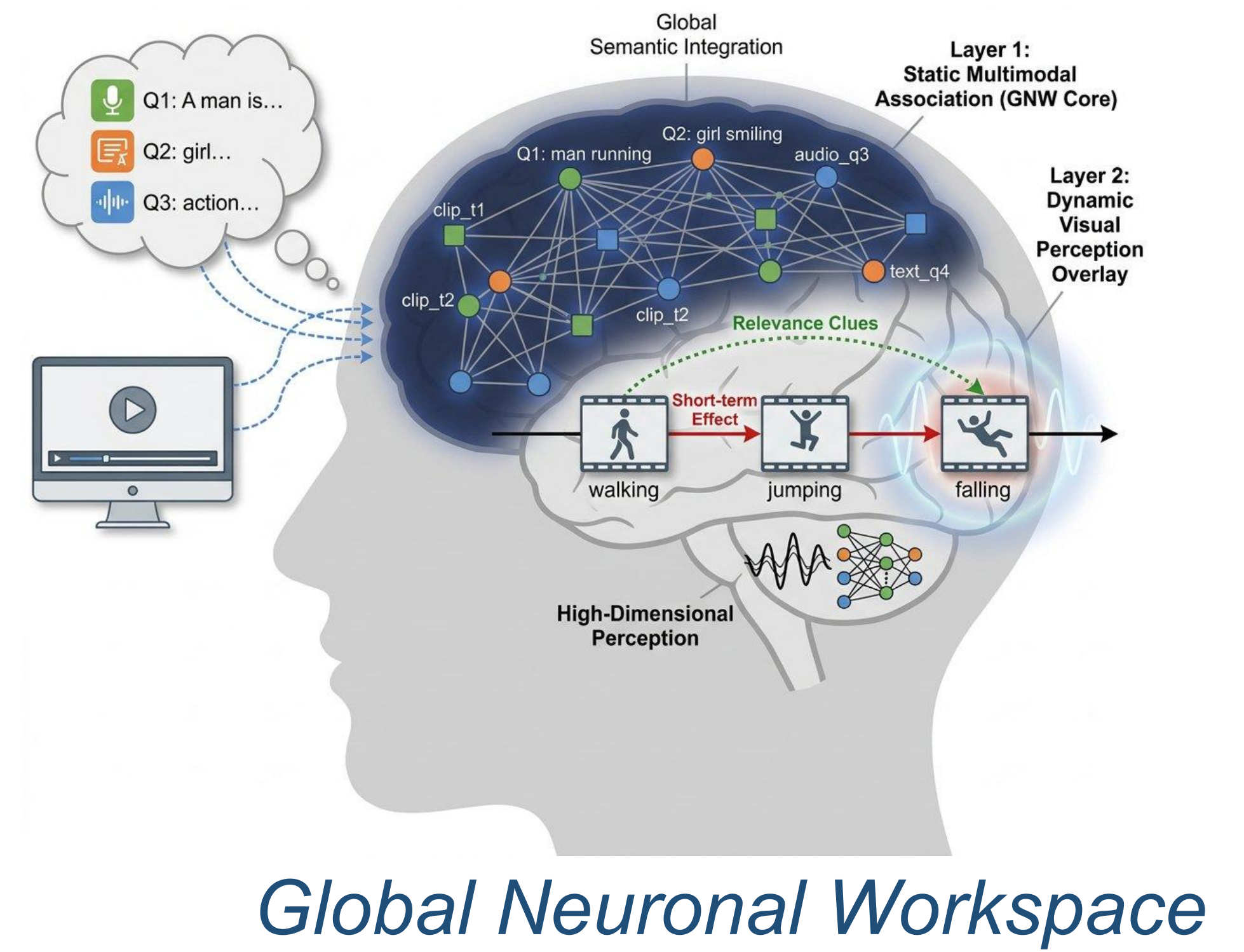
NLMR (Text Query): the man then rides the bike and it moves across the snow."

SLMR (Audio Query):



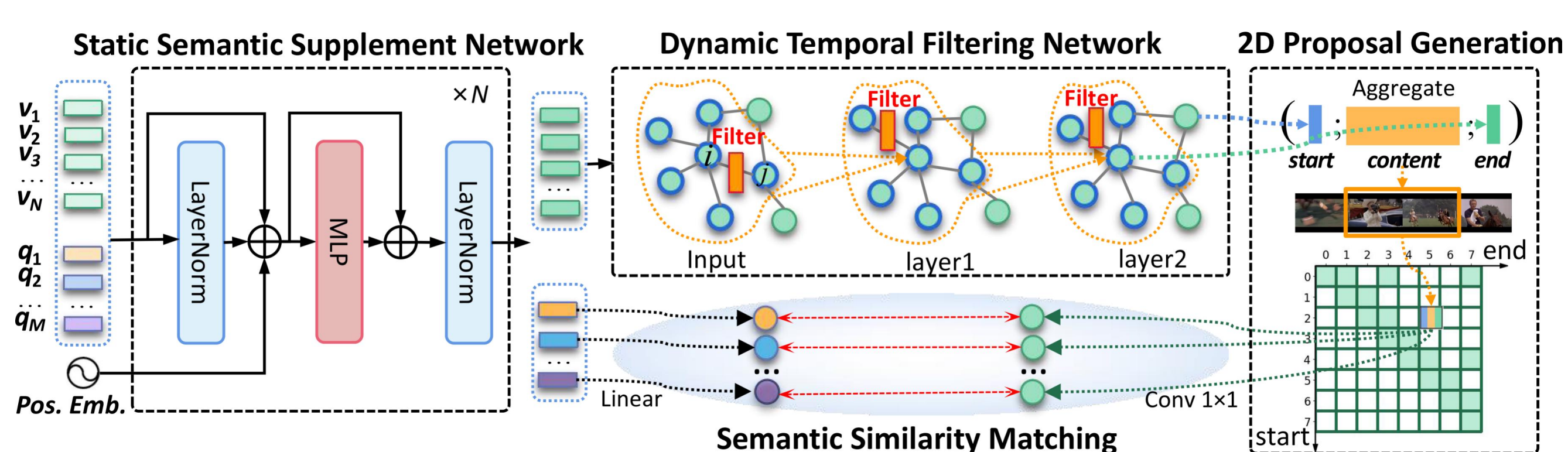
Ground Truth (GT): 80.75s | 147.34s

How can we understand videos like *humans* by combining **global cross-modal semantics** with perception?

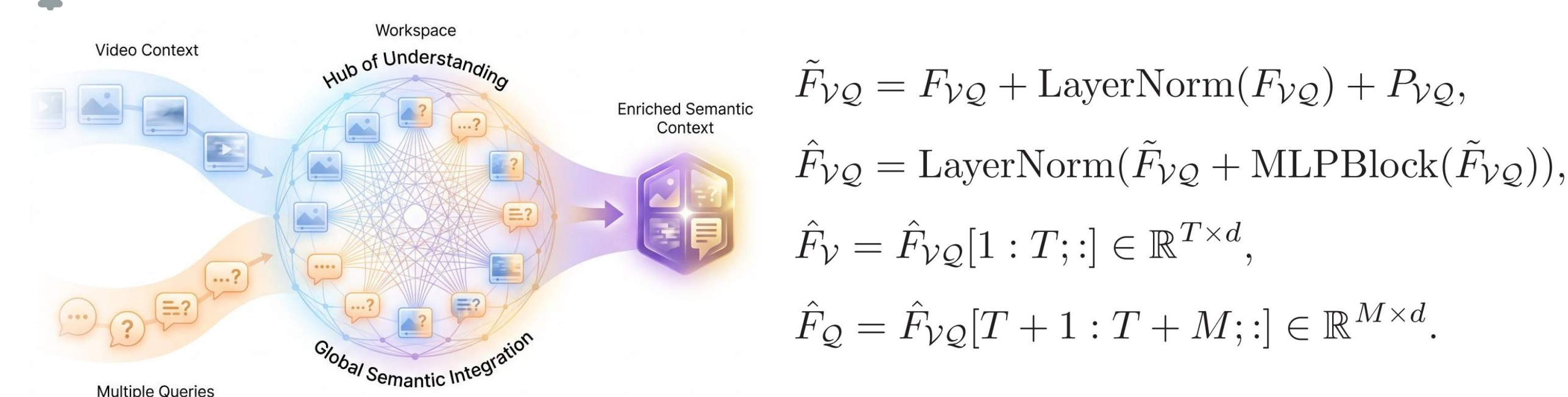


Method

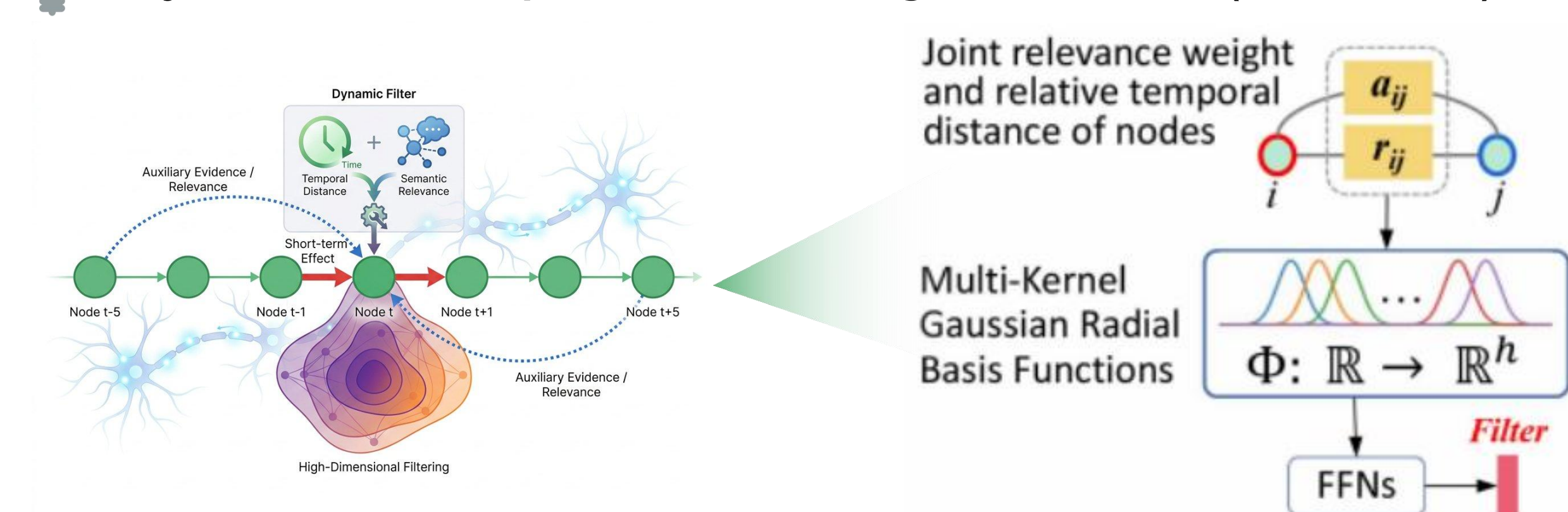
Unified Static and Dynamic Network (UniSDNet)



Static Semantic Supplement Network (*S³Net*)



Dynamic Temporal Filtering Network (*DTFNet*)

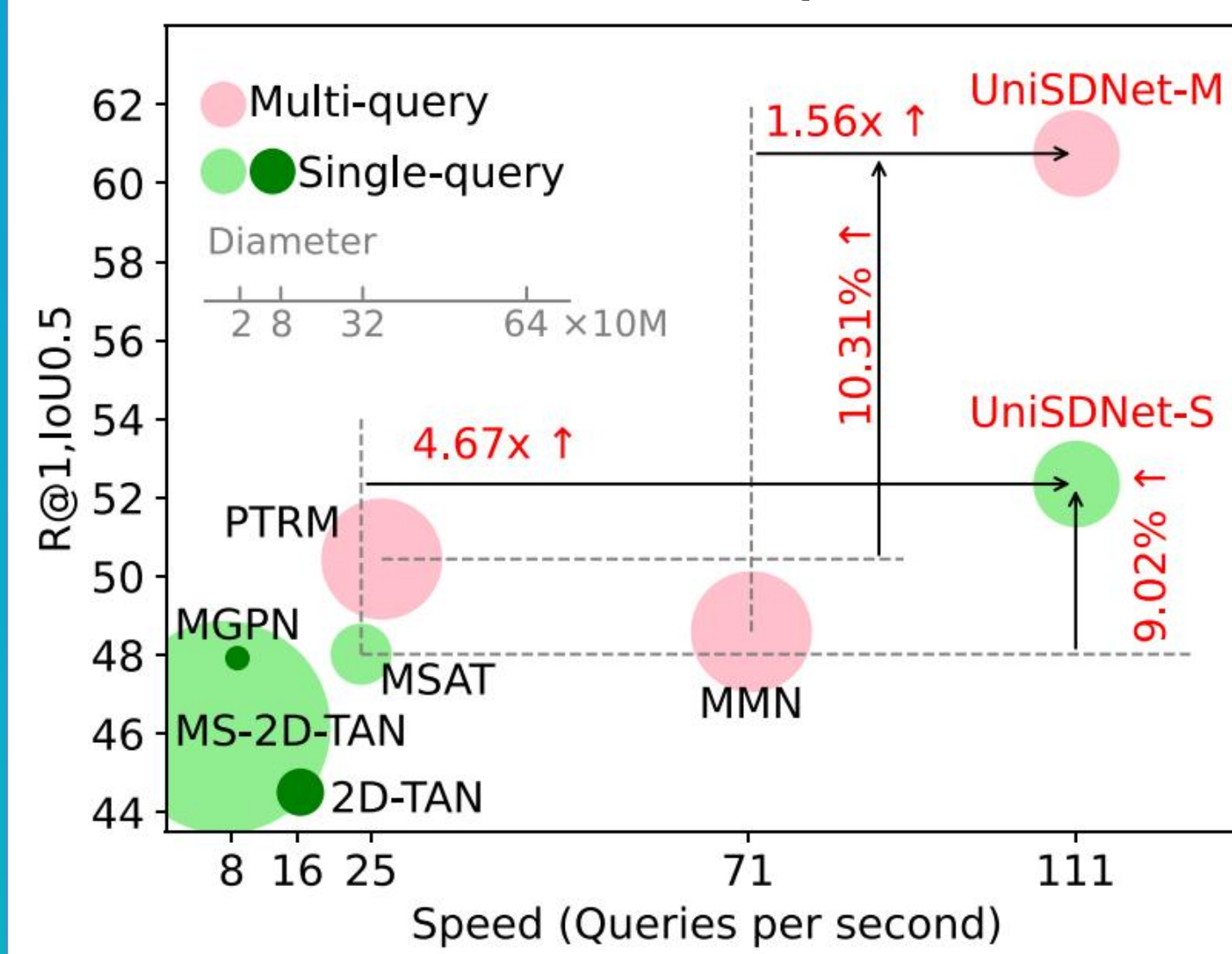


Results

Datasets: ANetCap\TACoS\Charades-STA\ANetSpeech\QVHighlights

Evaluation metrics: Recall\mIoU\mean average precision (mAP)

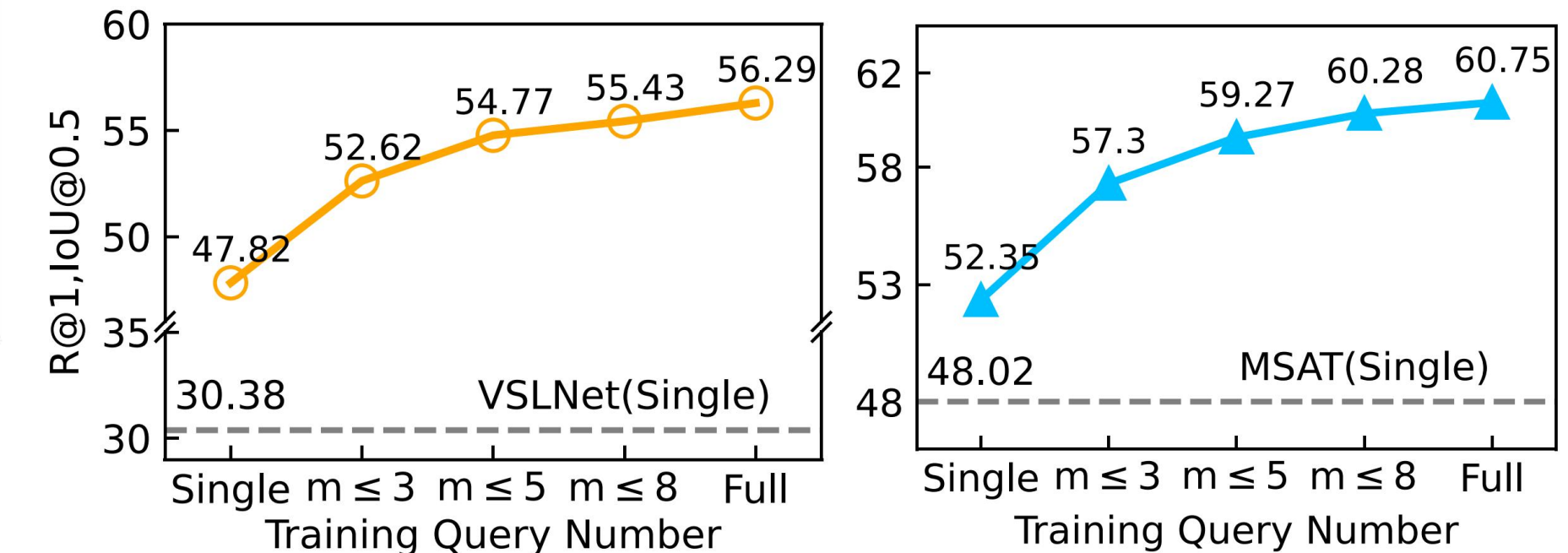
State-of-the-art performance



Module accuracy and efficiency

Method	Infer. Speed (s/query)	R@1, IoU@			mIoU
		0.3	0.5	0.7	
Transformer [28]	0.024	75.17	59.98	38.38	54.97
S4 [69]	0.030	70.41	55.11	34.93	51.40
Our S ³ Net	0.009	75.85	60.75	38.88	55.47

Query number ablation



Qualitative Results



- Achieves **SOTA** performance on both NLVG and SLVG benchmarks.
- Delivering $1.56 \times$ **faster** inference speed than baselines.

References

- [1] Jingjing Hu et, al. Unified Static and Dynamic Network: Efficient Temporal Filtering for Video Grounding, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 34, 2025.
- [2] Jingjing Hu et, al. Maskable Retentive Network for Video Moment Retrieval, *Proceedings of the 32nd ACM International Conference on Multimedia*, pp. 1476-1485, 2024.
- [3] Jingjing Hu et, al. Charades-STA Speech dataset is available at <https://zenodo.org/records/8019213>.

E-MAIL: xianhjj623@gmail.com

